

Exercise: Deleting SQL

Objective

Practice safely deleting records in a relational database, both individually and in bulk, while understanding the impact of foreign key constraints.

Environment

1. Create a new PostgreSQL database named `productdb`.
2. Run all setup scripts in this exercise against the `productdb` database.

Setup Script (Run First)

SQL

```
CREATE TABLE ProductCategory (  
    CategoryId INT PRIMARY KEY,  
    CategoryName VARCHAR(50)  
);  
  
CREATE TABLE Product (  
    ProductId INT PRIMARY KEY,  
    ProductName VARCHAR(100),  
    CategoryId INT,  
    Price DECIMAL(10, 2),  
    FOREIGN KEY (CategoryId) REFERENCES ProductCategory(CategoryId)  
);  
  
INSERT INTO ProductCategory VALUES  
(1, 'Books'),  
(2, 'Stationery'),  
(3, 'Clearance');  
  
INSERT INTO Product VALUES  
(101, 'Notebook', 2, 5.00),  
(102, 'Pen Set', 2, 7.50),  
(103, 'Calendar 2023', 2, 9.99),  
(104, 'Mystery Novel', 1, 15.00),  
(105, 'Classic Fiction', 1, 12.50),  
(106, 'Sticker Pack', 3, 2.99);
```

Instructions

Part 1: Delete a Single Record

1. Write a SELECT query to find the product with the name 'Calendar 2023'.
2. Use a DELETE statement to remove that product from the table.

Part 2: Delete Multiple Records

3. Write a SELECT query to preview all products in the 'Stationery' category.
4. Use a DELETE statement to remove all products in that category.

Part 3: Handle Foreign Key Dependencies

5. Try to delete the 'Books' category from the ProductCategory table.
 - What happens? Record the error message (if any).
6. Now safely remove the 'Books' category by deleting its related products first.

Part 4: Bonus Challenge

7. Write a DELETE statement that removes any category from ProductCategory only if no products are assigned to it.

Reflection Questions

- Why is it important to use SELECT before DELETE?
- What happens if you try to delete a parent record that is referenced by a foreign key?
- How would you protect critical records in a production environment?